

Transactions

Safe assets function like money for carrying out transactions.⁴⁴ The word central bankers use for this is *liquidity* (which is different from the notion of an illiquidity premium, discussed in chapter 13).⁴⁵ A major aim of monetary policy during the financial crisis, at least in the United States, was to alleviate liquidity shortages (see chapter 9). If we were to rank countries whose debt is the most useful in facilitating transactions to the least, it would be United States, and then Norway and Brazil again, with the order of the latter two depending on circumstances.

Another related role of risk-free assets, particularly currency (the sovereign bond with an instantaneous maturity), is a numeraire. We like to think of values as being independent of the units we use to measure them (your height is the same whether you measure it in inches or centimeters, for example), and numeraires transfer values from one unit to another. However, the choice of numeraire matters. Norway can report its return in local currency, U.S. dollars, or a currency basket. Each choice has consequences for the fund's volatility and perceived risk by the public.

Macro Factors

Chapter 9 discusses in detail how economic growth and inflation influence interest rates and hence safe asset prices. High economic growth makes countries more likely to honor their commitments, all else equal. Ranking countries from high growth to low, we have Brazil and then United States and Norway. If we order our countries from low to high inflation, with low inflation better for safe asset investors but high inflation better for investors with an eye to investment returns (see below), we have United States, Norway, and then Brazil.

Reserve Status

"Exorbitant privilege" was the phrase used by Valéry Giscard d'Estaing, the French finance minister in 1960, for the role of the U.S. dollar as a reserve currency and the ability of the U.S. to print all the dollars it wanted. The U.S. dollar is still the world's leading reserve asset, and our other two countries would then rank Norway and, as a distant third, Brazil.

Caballero and Farhi (2013) argue that demand for reserve assets, especially U.S. sovereign bonds, has been steadily increasing. The supply of safe U.S. assets is limited by the health of the federal budget and the fact that the United States is shrinking as a proportion of the world economy. Our exorbitant privilege is that we can borrow cheaply through large reserve demand among foreigners, especially in emerging markets. As crises erupt, the U.S. government is likely to encounter

⁴⁴This literature dates back to Sidrauski (1967) and Tobin (1969). A recent example is Krishnamurthy and Vissing-Jørgensen (2012).

⁴⁵Important early references are Woodford (1990) and Holmstrom and Tirole (1998). There is a huge resurgence of papers on this topic after the financial crisis.